Main.java

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```java
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = {10, 20, 30, 40, 50};
        System.out.print("Enter the index: ");
        int index = sc.nextInt();
        if (index >= 0 && index < arr.length) {
            System.out.println("Element at index " + index + " is: " + arr[index]
                );
        } else {
            System.out.println("Invalid index");
        }
        sc.close();
    }
}
```

Output

```
Enter the index: 2
Element at index 2 is: 30

=== Code Execution Successful ===
```

```java
public class Main {
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40, 50};
        int key = 30;
        int low = 0, high = arr.length - 1;
        while (low <= high) {
            int mid = (low + high) / 2;
            if (arr[mid] == key) {
                System.out.println("Element found at index: " + mid);
                return;
            } else if (arr[mid] < key) {
                low = mid + 1;
            } else {
                high = mid - 1;
            }
        }
        System.out.println("Element not found");
    }
}
```

```
Element found at index: 2

=== Code Execution Successful ===
```

```java
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        int max = arr[0];
        for (int i = 1; i < n; i++) {
            if (arr[i] > max) {
                max = arr[i];
            }
        }
        System.out.println("Maximum element = " + max);
    }
}
```

```
Enter the number of elements: 5
Enter the array elements:
12 45 8 67 23
Maximum element = 67

=== Code Execution Successful ===
```

```java
import java.util.Arrays;
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.print("Enter value of k: ");
        int k = sc.nextInt();
        Arrays.sort(arr);
        System.out.println("The " + k + "th smallest element is: " + arr[k - 1]);
    }
}
```

```
Enter number of elements: 5
Enter array elements:
7 2 9 4 1
Enter value of k: 3
The 3th smallest element is: 4

=== Code Execution Successful ===
```

```java
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter the number of elements: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter the array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println("All possible pairs are:");
        for (int i = 0; i < n; i++) {
            for (int j = i + 1; j < n; j++) {
                System.out.println("(" + arr[i] + ", " + arr[j] + ")");
            }
        }
        sc.close();
    }
}
```

```
Enter the number of elements: 4
Enter the array elements:
1 2 3 4
All possible pairs are:
(1, 2)
(1, 3)
(1, 4)
(2, 3)
(2, 4)
(3, 4)

=== Code Execution Successful ===
```